One ejection, observed through real arterial paths

Forward wave

$f(t)$ at the aortic root

Path, not a free delay

$\tau_s = L_s / v$

L_s from height

Speed stays in the measured range

order follows the arteries

neck before face, both before the palm

What a site is allowed to show

$$\hat{y}_s(t) = a_s f(t - \tau_s) + b_s f(t - \tau_s - \delta_s)$$

second term is the reflection; it grows toward the periphery

Timing uses the foot

not the systolic peak

the peak is moved by the reflection

face regions are too close

All observed sites fit this same ejection

A missing site removes one equation. The ejection, the lengths, and the speed stay.

Confidence falls when fewer sites agree, or when speed would have to leave the physiological range.

Delays from the central aorta, not the QRS: carotid about 30 ms, brachial about 60 ms, radial about 80 ms, femoral about 75 ms.